

POTAPOVO, Russian Federation

WMO: 231740

Lat: 68.681N

Lon: 86.273E

Elev: 41

StdP: 100.83

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.2	-45.7	-50.2	0.0	-48.0	-47.8	0.0	-45.6	18.2	-22.7	16.0	-24.4	0.6	180	0.844

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	25.7	17.4	22.9	16.5	20.6	15.4	18.6	23.5	17.3	21.7	16.0	19.8	3.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.0	20.2	15.5	11.0	19.7	13.9	10.0	18.1	53.0	23.1	48.9	21.8	44.8	20.0	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	9.8	8.5	-48.5	27.4	2.3	3.4	-50.2	29.9	-51.6	31.9	-52.9	33.8	-54.6	36.2		
(5)			-49.0	19.8	2.7	2.4	-51.0	21.6	-52.6	23.0	-54.1	24.4	-56.1	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.2	-28.7	-25.4	-18.1	-12.2	-2.5	9.2	14.5	10.6	4.8	-6.5	-21.5	-23.4	(6)
(7)		DBStd	16.62	10.34	9.46	9.62	7.97	5.27	6.12	4.51	3.71	4.50	6.77	8.91	10.31	(7)
(8)		HDD10.0	6911	1201	991	871	666	388	88	11	37	166	512	946	1035	(8)
(9)		HDD18.3	9695	1459	1224	1129	916	647	278	136	239	407	770	1196	1293	(9)
(10)		CDD10.0	280	0	0	0	0	0	65	151	56	8	0	0	0	(10)
(11)		CDD18.3	23	0	0	0	0	0	6	17	1	0	0	0	0	(11)
(12)		CDH23.3	180	0	0	0	0	0	42	137	1	0	0	0	0	(12)
(13)		CDH26.7	28	0	0	0	0	0	6	22	0	0	0	0	0	(13)
(14)	Wind	WSAvg	3.5	3.7	3.9	3.3	3.4	3.5	3.4	3.1	3.1	3.4	3.7	3.4	3.9	(14)
(15)	Precipitation	PrecAvg	871	72	64	62	62	61	59	61	80	69	110	85	86	(15)
(16)		PrecMax	1110	145	120	131	121	105	105	160	134	143	178	163	143	(16)
(17)		PrecMin	431	13	13	11	3	12	8	10	29	30	33	18	8	(17)
(18)		PrecStd	139	34	28	30	27	22	25	33	24	25	37	36	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.1	-5.3	0.7	7.1	12.4	27.7	28.9	22.9	18.3	7.9	-1.3	-4.0	(19)
(20)			MCWB	-7.7	-5.8	-0.2	2.8	7.0	18.4	18.7	17.3	12.4	7.0	-2.0	-4.7	(20)
(21)		2%	DB	-10.4	-7.2	-0.7	3.7	9.2	24.5	27.0	20.5	15.1	4.9	-3.3	-6.0	(21)
(22)			MCWB	-10.9	-7.6	-1.7	1.2	5.3	16.1	18.6	15.7	11.4	4.1	-3.8	-6.6	(22)
(23)		5%	DB	-12.6	-10.0	-2.2	1.0	6.2	21.5	24.7	18.5	13.1	3.0	-5.3	-7.4	(23)
(24)			MCWB	-13.0	-10.4	-2.9	-0.6	3.4	14.7	17.3	14.6	10.2	2.2	-5.8	-7.9	(24)
(25)		10%	DB	-14.8	-12.5	-4.2	-0.7	4.2	18.7	22.2	16.6	11.0	1.2	-8.6	-9.5	(25)
(26)			MCWB	-15.3	-12.9	-4.9	-1.9	1.9	13.5	16.5	13.7	8.7	0.3	-9.1	-10.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.8	-5.6	-0.1	3.4	7.3	18.7	21.1	18.6	14.0	7.2	-1.6	-4.6	(27)
(28)			MCDWB	-7.6	-5.3	0.5	6.3	11.9	24.9	26.0	21.9	16.3	8.3	-1.1	-4.2	(28)
(29)		2%	WB	-10.9	-7.6	-1.6	1.3	5.3	17.0	19.3	16.7	12.1	4.3	-3.6	-6.5	(29)
(30)			MCDWB	-10.6	-7.3	-0.9	3.6	8.7	22.3	24.6	19.3	14.7	4.9	-3.1	-6.1	(30)
(31)		5%	WB	-13.0	-10.4	-3.0	-0.5	3.7	15.5	18.2	15.2	10.4	2.2	-5.8	-7.8	(31)
(32)			MCDWB	-12.7	-10.0	-2.1	0.8	6.1	20.5	22.9	17.7	12.9	2.8	-5.3	-7.4	(32)
(33)		10%	WB	-15.3	-12.9	-4.8	-1.9	2.1	13.9	17.2	13.9	8.9	0.6	-9.1	-9.9	(33)
(34)			MCDWB	-14.9	-12.5	-4.2	-0.6	3.8	18.6	21.5	16.4	10.5	1.2	-8.6	-9.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.1	7.7	9.7	9.5	6.3	7.6	8.3	6.5	5.3	5.3	7.6	7.8	(35)
(36)			MCDBR	8.7	8.8	7.8	9.5	7.9	11.2	12.1	9.2	7.6	4.5	8.3	7.8	(36)
(37)			MCWBR	8.6	8.4	7.4	8.1	5.2	6.0	6.0	5.5	5.0	4.1	8.3	7.4	(37)
(38)		5% WB	MCDBR	8.7	8.0	8.2	9.1	7.5	10.5	10.0	7.5	6.6	4.3	8.9	7.7	(38)
(39)			MCWBR	8.6	7.9	7.8	8.1	5.0	6.2	5.9	5.3	4.9	4.0	8.8	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.216	0.253	0.305	0.328	0.354	0.369	0.404	0.393	0.328	0.255	0.199	N/A	(40)	
(41)		taud	1.857	1.971	2.005	2.050	2.123	2.378	2.285	2.272	2.407	2.109	1.885	N/A	(41)	
(42)		Ebn at Noon	38	487	687	798	823	828	774	732	701	488	37	0	(42)	
(43)		Edh at Noon	7	50	92	119	125	101	108	97	65	41	5	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.02	0.43	1.98	4.12	5.64	5.41	5.04	3.13	1.65	0.54	0.07	0.00	(44)	
(45)		RadStd	0.00	0.05	0.21	0.33	0.45	0.49	0.66	0.36	0.17	0.10	0.01	0.00	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page